ASSIGNMENT

Course Code : ICE 2341 **Course Title :** Discrete Mathematics

Title/Topic : Course Final Assignment

No.	Evaluation Criteria with Marks	Put Tick (✓) Mark					Marks Obtained	Remarks
		Excellent	Good	Fair	Poor	Fail		
ASSIGNMENT								
1	Idea with Focus (1)							
2	Organization (1)							
3	Content (2)							
4	Time Management (1)							
	TOTAL							
Date of Submission : 07th August 2026								Teacher's Signature

Semester : Summer **Year :** 2026 **Level-Term :** L2-T3 **Section :** A

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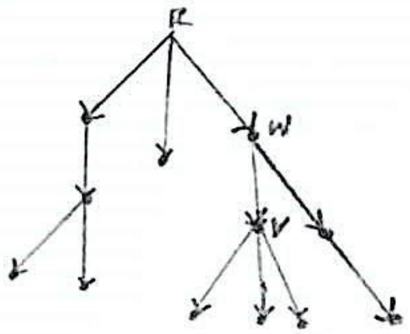
Q-1:

~~condition~~

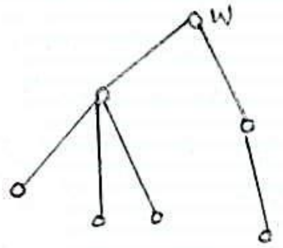
Consider the following rooted tree T:

1. How many children has vertex v?
2. Draw the subtree of vertex w?
3. What is the depth of vertex v?
4. What is the height of the tree?
5. Who is the parent of w?



⇒

1. children of v = 3
2. subtree of vertex w,

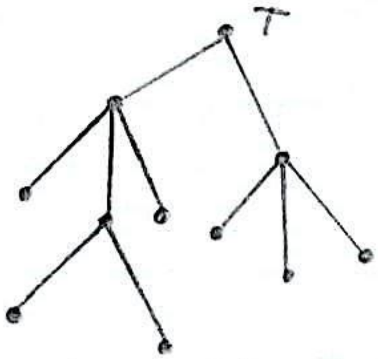


3. Depth of vertex, v = 2
4. Height of the tree = 3
5. Parent of w = r .

Q-2:

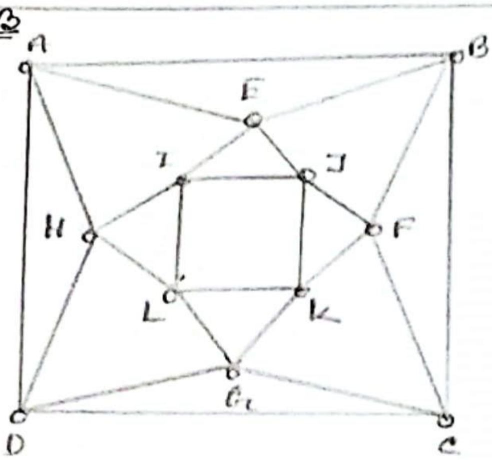
Give an example of 3-ary tree which is not a full 3-ary tree.

⇒

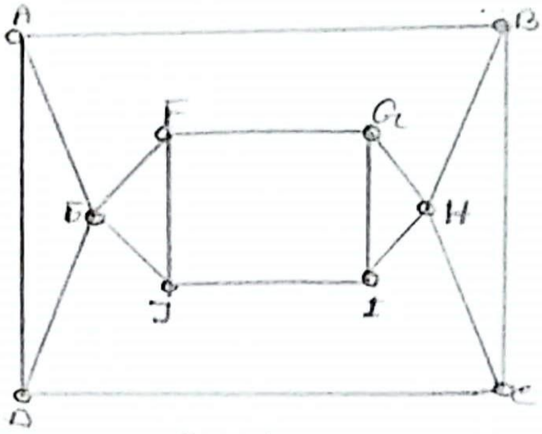


It's a 3-ary tree but not full 3-ary tree.

Q-3



Graph-1



Graph-2

1. Find a Hamiltonian Path in each graph.
2. Find an Eularian path or cycle in each graph. If it is not possible, explain why?

⇒

1. Hamiltonian path for each graph,

Graph-1 :- A, B, C, D, G, K, F, J, E, I, L, H

Graph-2 :- A, B, C, D, E, F, G, H, I, J

2. Eularian path in each graph,

For graph 1 -

A, B, C, D, G, C, F, B, E, J, F, K, J, I, E, A, H, I, K, G, L, H, D, A.

As each of the vertex has even degree (4).

For Graph 2 -

No Eularian path or circuit exist. As -

i. A, B, C, D, F, G, J, I = Degree 3 each.

ii. E, H = Degree 4 each.

That gives 8 vertics of odd degree. Eular's theorem requires 0 odd-degree vertex for a circuit. Also exactly 2 odd-degree vertex for a path.

So, graph 2 doesn't have a Eular path or cycle.

Q-4:

Let, A = {1, 2, 3, 4, 5} Give an example of a relation on A Which -

1. is transitive
2. is transitive but not reflexive
3. is reflexive but not symmetric
4. is antisymmetric.

List the elements, draw the digraph and matrix reppreshting the relations.

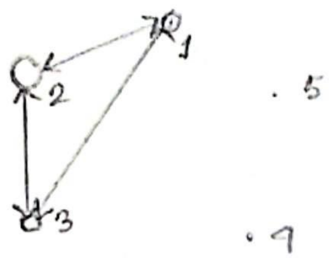
⇒

1. Transitive –

$R_1 = \{(1,1), (1,2), (1,3), (2,2), (2,3)\}$

$$\therefore M_{R_1} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Digraph,

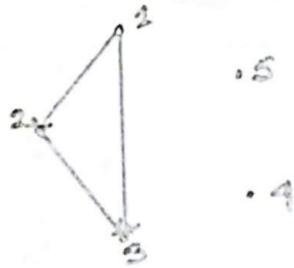


2. Transitive but not reflexive,

$R_2 = \{(1,2), (1,3), (2,3)\}$

$$M_{R_2} = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Digraph,

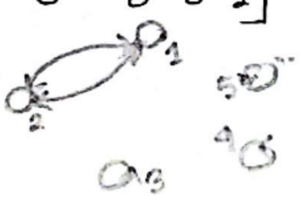


3. Reflexive but not symmetric,

$R_3 = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1)\}$

$$M_{R_3} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Digraph –

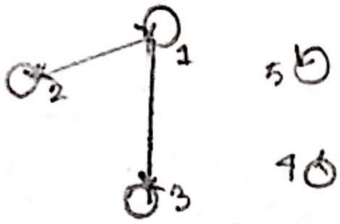


4. Antisymmetric —

$R_4 = \{(1,1),(1,2),(1,3),(2,2),(3,3),(4,4),(5,5)\}$

$$M_{R_4} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Digraph —



Q-5: Draw the graph G with adjacency matrix M. and give the partition of its vertices into connected coponents.

$$M = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

⇒ From matrix M.

The set of vertex is, $V = \{1, 2, 3, 4, 5, 6\}$

So, Adjacency List,

Vertex	Adjacent Vertics
1	3, 6
2	
3	1, 6
4	5
5	4
6	1, 3

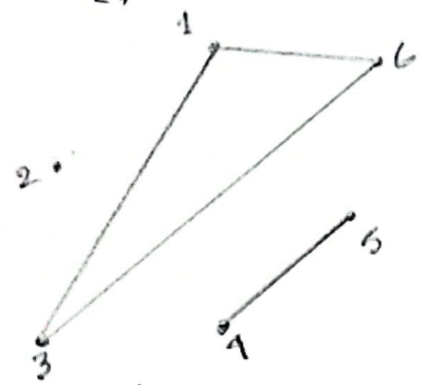
Adjacency matrix,

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

So, the edges are, $= \{(1,3), (1,6), (3,1), (3,6), (4,5), (5,4), (6,1), (6,3)\}$

$= \{(1,3), (1,6), (3,6), (4,5)\}$

∴ Graph G,



~~so connetected~~

so connectcd component,

$\{1, 3, 3\}, \{4, 5\}, \{2\}$

∴ graph edges $= \{(1,3), (1,6), (3,6), (4,5)\}$

connected components $= \{\cancel{1}\ 1, 3, 6\}, \{4, 5\}, \{2\}$